



gRPC: Conhecendo novo modelo de comunicação entre microserviços

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Agenda

01. Microserviços

O que é? Conceitos. Comunicação síncrona e assíncrona.

02. HTTP/2

O que é? Principais benefícios e desvantagens.

03. Protocol buffers aka Protobuf

O que é? Vantagens em relação a outros formatos.

04. gRPC

O que é? Conceitos. Por que eu gostaria de usar o gRPC? Demo .

Microserviços

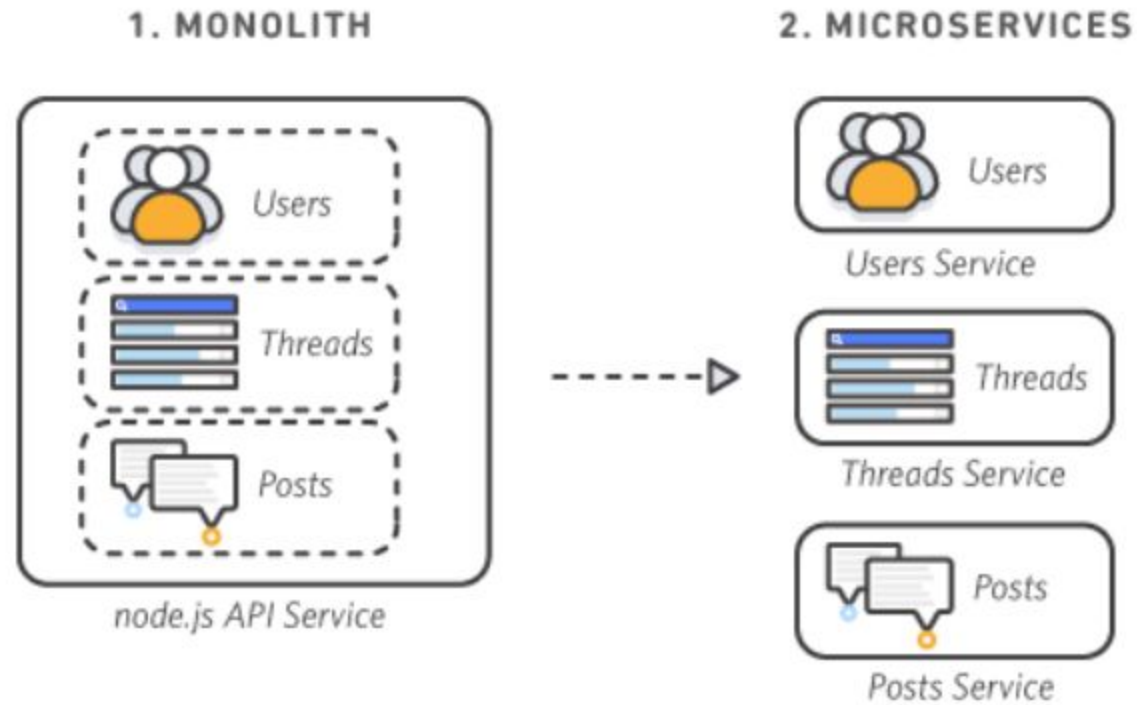
O que é? Conceitos. Comunicação síncrona e assíncrona.
Exemplos na GVP.



Permite usar a melhor **ferramenta**
para resolver seu **problema** de forma
distribuída.



Monolítico vs Microserviços



Características comuns das arquiteturas de microserviços



Tecnologia heterogênea



Resiliência



Escalabilidade



Facilidade de deploy



Alinhamento organizacional

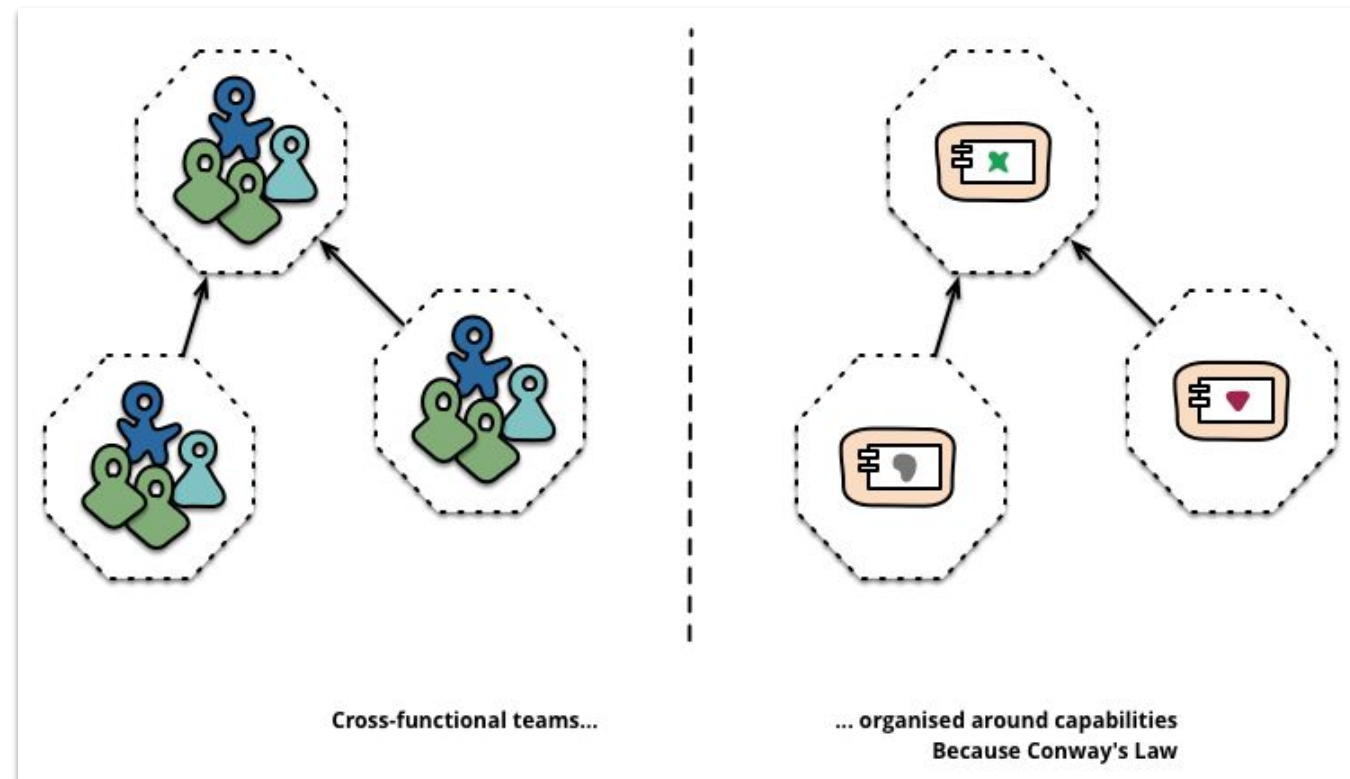
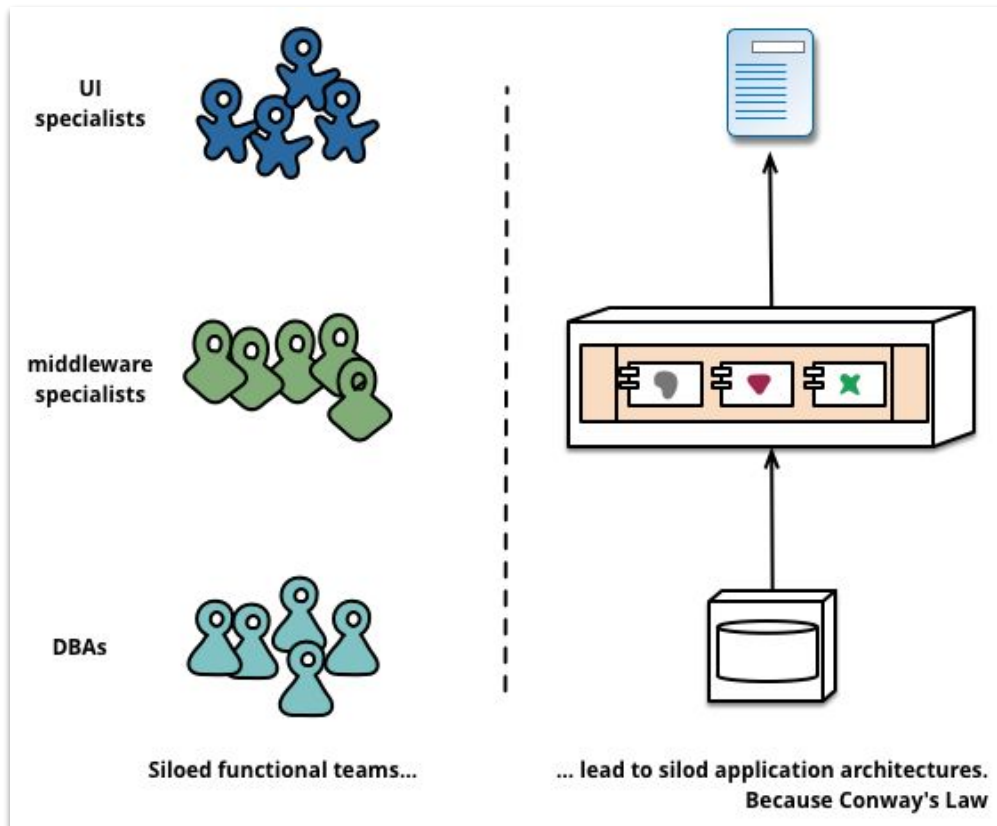


Pequeno e focado

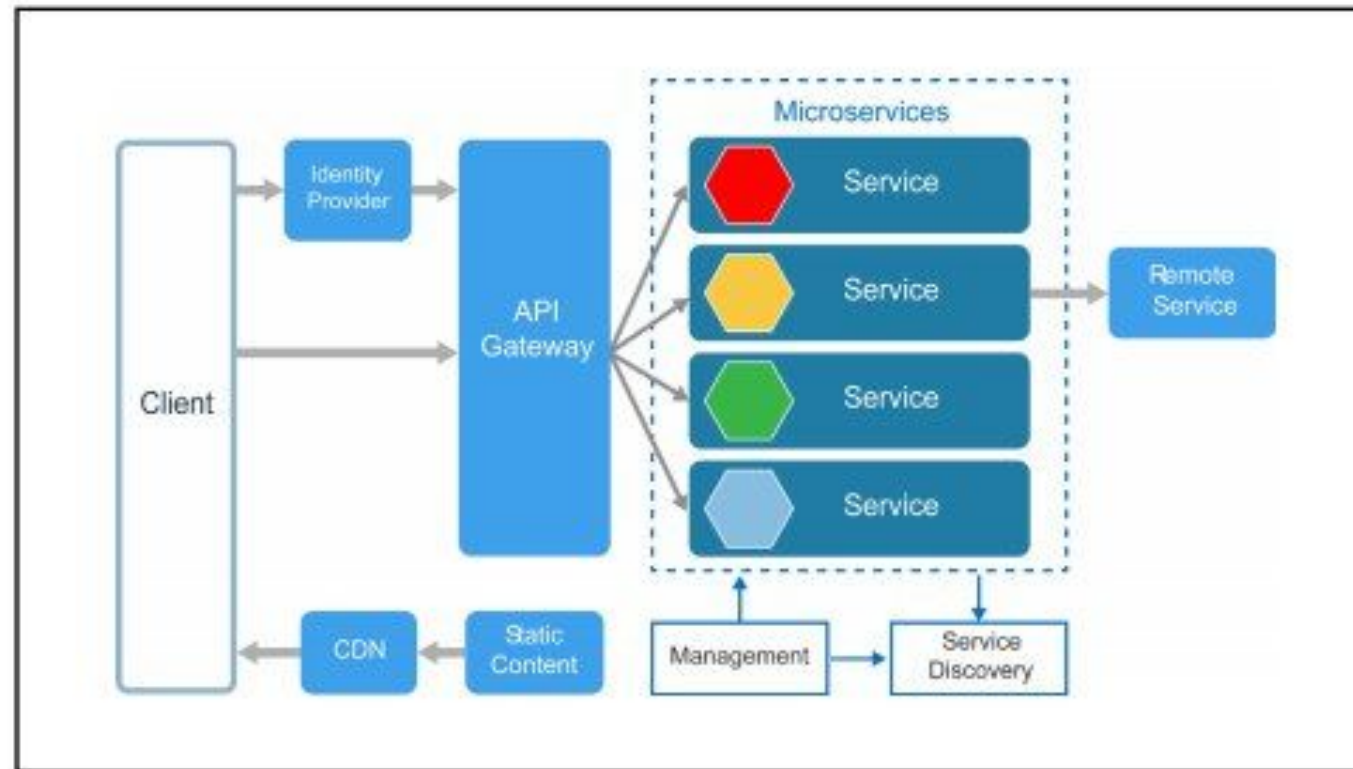
Lei de Conway

Qualquer empresa que projeta um sistema, inevitavelmente, produz um projeto cuja a estrutura é uma cópia da estrutura de comunicação da organização.

[Melvin Conway \(1967\)](#)

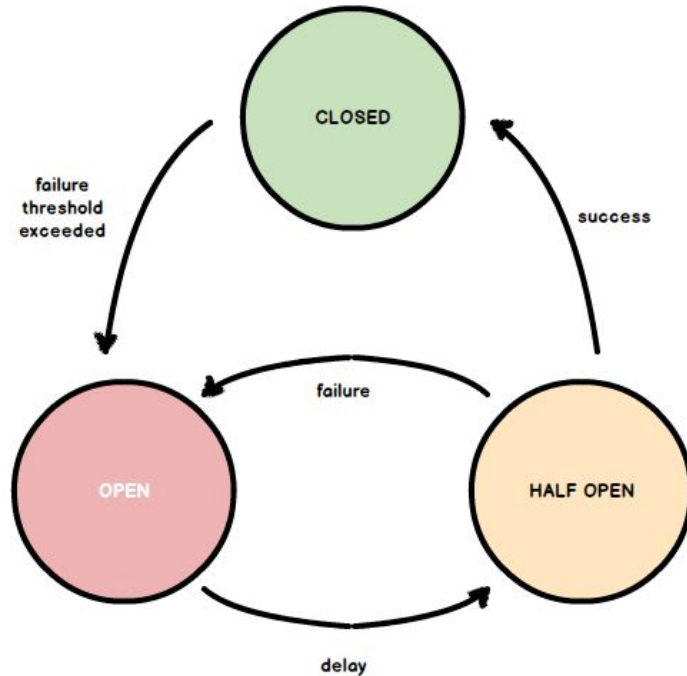


Padrões de design dos microserviços



<https://azure.microsoft.com/en-in/campaigns/cloud-application-architecture-guide/>

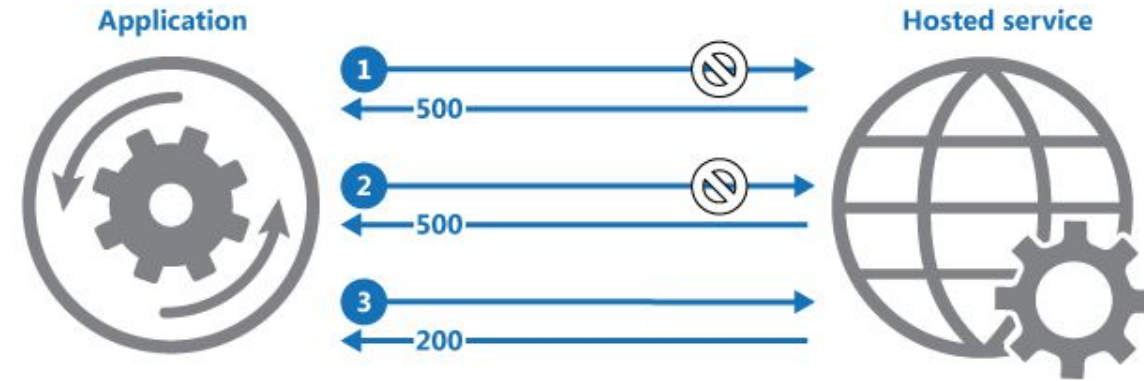
Comunicação síncrona



<https://github.com/App-vNext/Polly/wiki/Circuit-Breaker>



<http://www.thepollyproject.org/2019/06/27/simmy-the-monkey-for-making-chaos/>



- 1: Application invokes operation on hosted service. The request fails, and the service host responds with HTTP response code 500 (internal server error).
- 2: Application waits for a short interval and tries again. The request still fails with HTTP response code 500.
- 3: Application waits for a longer interval and tries again. The request succeeds with HTTP response code 200 (OK).

<https://code.4noobz.net/retry-pattern-with-polly/>

Comunicação assíncronas



Entre outras mensagerias...

Principais gargalos em uma aplicação. (não só em microserviços)



Rede



I/O



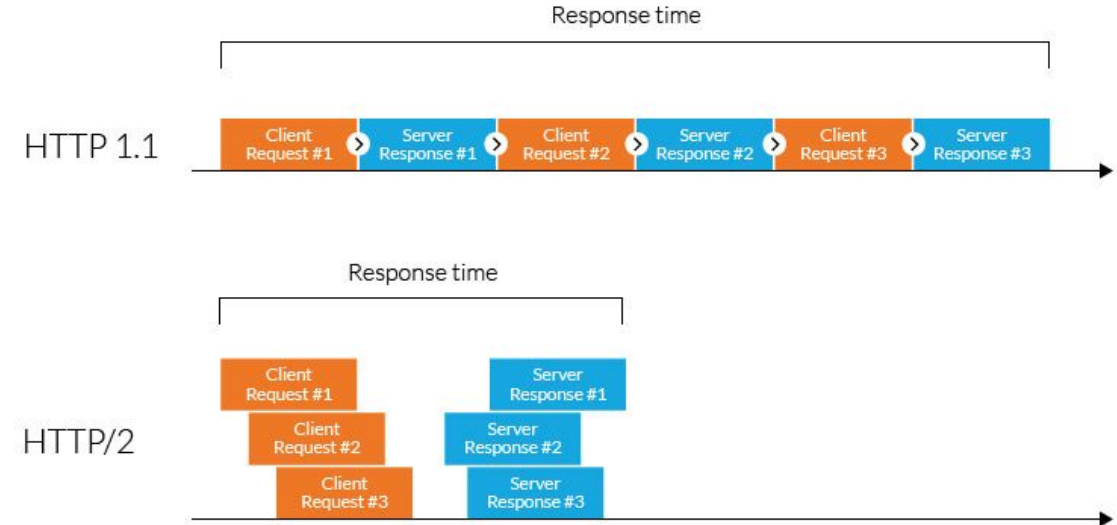
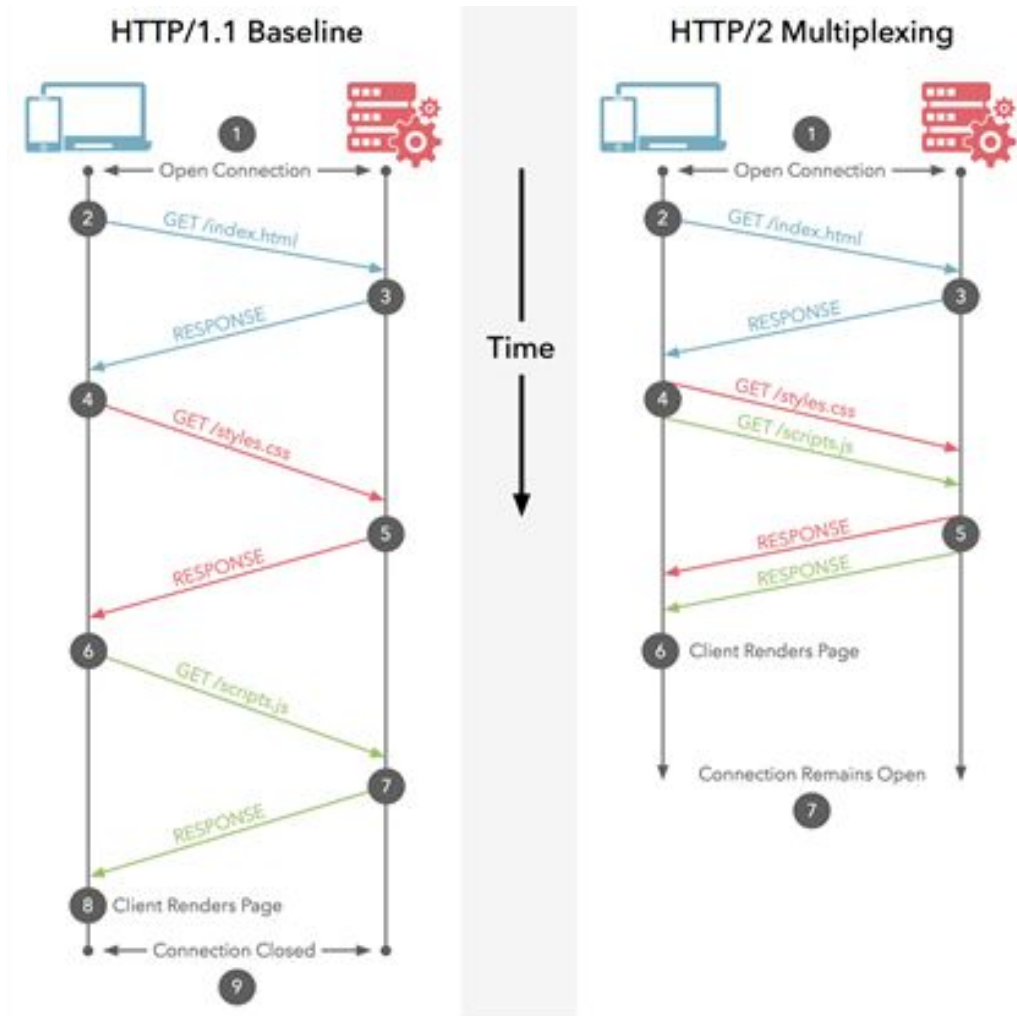
GC (Garbage Collector)

HTTP/2

O que é? Principais benefícios e desvantagens.



Conexões multiplexadas



<https://www.imperva.com/learn/performance/http2/>

Tudo é binário

HTTP/1.1 Headers vs HTTP/2 Headers

```
GET / HTTP/1.1
Host: www.devxperience.com.br
HTTP/1.1 200 OK
Content-Type: text/html; charset=utf-8
Content-Language: pt-br Date: Thu, 01 Set 2017 00
Cache-Control: private
```

189 bytes

```
?c?8)E8??%:?$
```

```
T?%.?JI?;1?R
,1?T?%.?JI?;1?R
?c?8)E8??%:?$
```

61 bytes

32% do
conteúdo original

Suporte



Firefox, Chrome, IE, Edge, Safari, Opera e os browsers mobile nas versões mais recentes.



O Apache suporta através do módulo *mod_h2*;



h2o, Tomcat, IIS, nginx e Node.js suportam na versão mais recentes;



Akamai, AWS CloudFront, Fastly e Cloudflare são os principais CDNs que suportam o HTTP/2.

Protocol buffers aka Protobuf



O que é? Vantagens em relação a outros formatos.

Protobuf



Protocol buffers são apenas um formato de intercâmbio.



Protocol Buffers é rápido e eficiente.

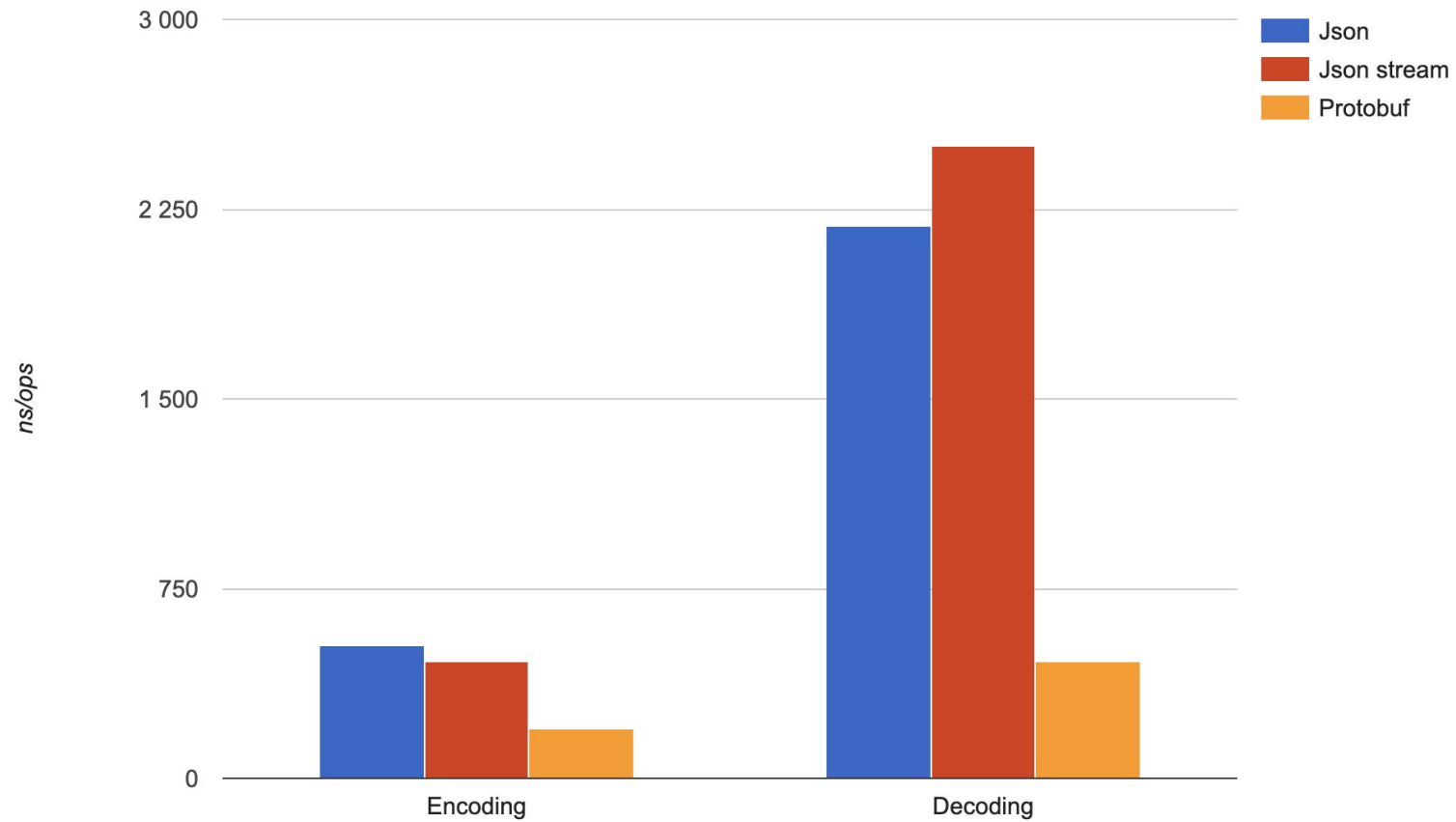


Começou inicialmente na Google em 2001 se tornou publica em 2008.



Com aproximadamente 568 contribuidores no projeto principal.

Protobuf e Json



gRPC



O que é? Conceitos. Por que eu gostaria de usar o gRPC?
Demo 🐱.



O que é? Conceitos.



gRPC é um moderno, open source remote procedure call (RPC) framework que pode ser executado em qualquer lugar. Ele permite que os aplicativos cliente e servidor se comuniquem de forma transparente.



gRPC usa HTTP/2 para transporte, Protocol Buffers como linguagem de descrição da interface e fornece recursos como autenticação, streaming bidirecional e controle de fluxo, etc...



gRPC está sobre a licença Apache 2.0 com aproximadamente 485 contribuidores no projeto principal.



gRPC foi originalmente desenvolvido internamente pela google atualmente é parte da Cloud Native Computing Foundation (CNCF).

Cloud Native Computing Foundation (CNCF)

Cloud Native Computing Foundation (CNCF) (29)							
 cloudevents CloudEvents Cloud Native Computing Foundation (CNCF) ★ 635	 CNI Container Network Interface (CNI) Cloud Native Computing Foundation (CNCF) ★ 1,543	 CSI Container Storage Interface (CSI) Cloud Native Computing Foundation (CNCF) ★ 261	 containerd Cloud Native Computing Foundation (CNCF) ★ 2,934	 CoreDNS Cloud Native Computing Foundation (CNCF) ★ 2,585	 cri-o Cloud Native Computing Foundation (CNCF) ★ 1,280	 envoy Cloud Native Computing Foundation (CNCF) ★ 6,557	 fluentd Cloud Native Computing Foundation (CNCF) ★ 6,766
 gRPC Cloud Native Computing Foundation (CNCF) ★ 17,148	 HARBOR Cloud Native Computing Foundation (CNCF) ★ 5,377	 Helm Cloud Native Computing Foundation (CNCF) ★ 7,610	 Jaeger Cloud Native Computing Foundation (CNCF) ★ 5,817	 kubernetes Cloud Native Computing Foundation (CNCF) ★ 41,500	 LINKERD Cloud Native Computing Foundation (CNCF) ★ 4,671	 minikube Cloud Native Computing Foundation (CNCF) ★ 10,801	 NATS Cloud Native Computing Foundation (CNCF) ★ 4,598
 Notary Cloud Native Computing Foundation (CNCF) ★ 1,516	 Open Policy Agent Cloud Native Computing Foundation (CNCF) ★ 1,426	 OPENMETRICS Cloud Native Computing Foundation (CNCF) ★ 326	 OPENTRACING Cloud Native Computing Foundation (CNCF) ★ 1,350	 Prometheus Cloud Native Computing Foundation (CNCF) ★ 18,927	 rkt Cloud Native Computing Foundation (CNCF) ★ 8,204	 ROOK Cloud Native Computing Foundation (CNCF) ★ 3,431	 spiffe Cloud Native Computing Foundation (CNCF) ★ 291
 SPIRE Cloud Native Computing Foundation (CNCF) ★ 261	 Telepresence Cloud Native Computing Foundation (CNCF) ★ 1,175	 The Update Framework (TUF) Cloud Native Computing Foundation (CNCF) ★ 785	 TiKV Cloud Native Computing Foundation (CNCF) ★ 3,713	 Vitess Cloud Native Computing Foundation (CNCF) ★ 6,539			

Por que eu gostaria de usar o gRPC? Os principais cenários de uso:



Sistemas distribuídos de baixa latência, altamente escaláveis.



Desenvolvimento de clientes móveis que estão se comunicando com um servidor em nuvem.



Projetando um novo protocolo que precisa ser preciso, eficiente e independente do idioma.



Design em camadas para permitir a extensão, por exemplo. autenticação, balanceamento de carga, registro e monitoramento etc.

Opções para comunicação entre serviços

REST (REpresentational State Transfer)

Foco nos Recursos

Adota a semântica do HTTP

Acoplamento solto

Mensagens frequentemente baseadas em texto

RPC (Remote Procedure Call)

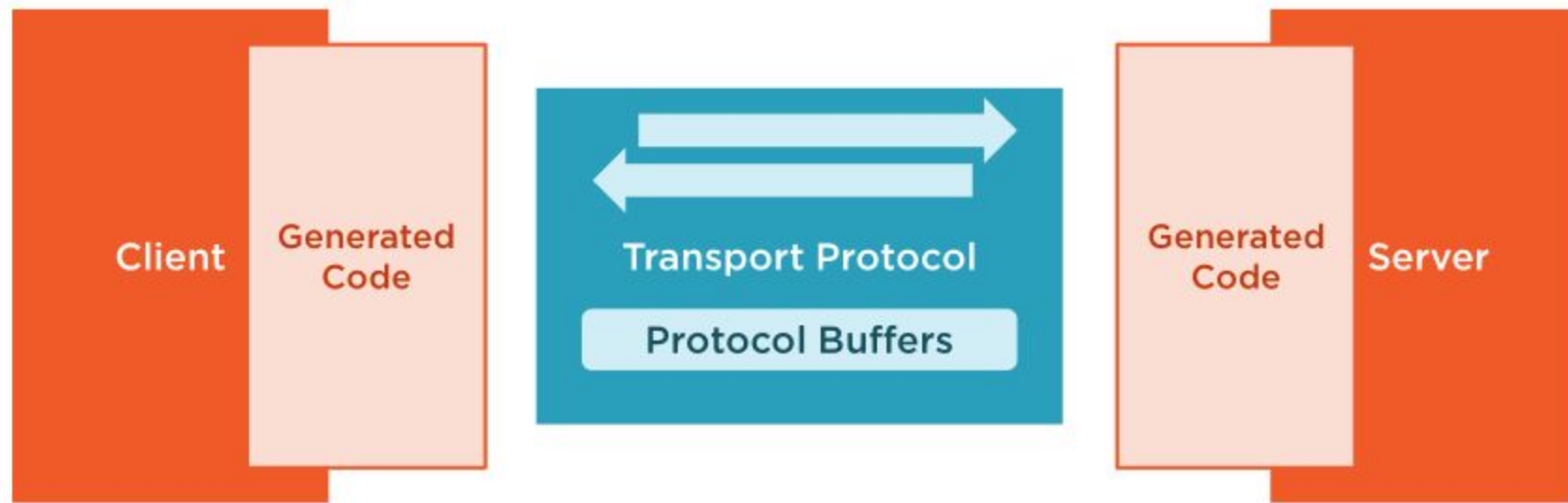
Foco nas Ações

Adota a semântica de programação

Acoplamento mais apertado

Mensagens frequentemente baseadas em binários

Estrutura de gRPC



Tipos de mensagem

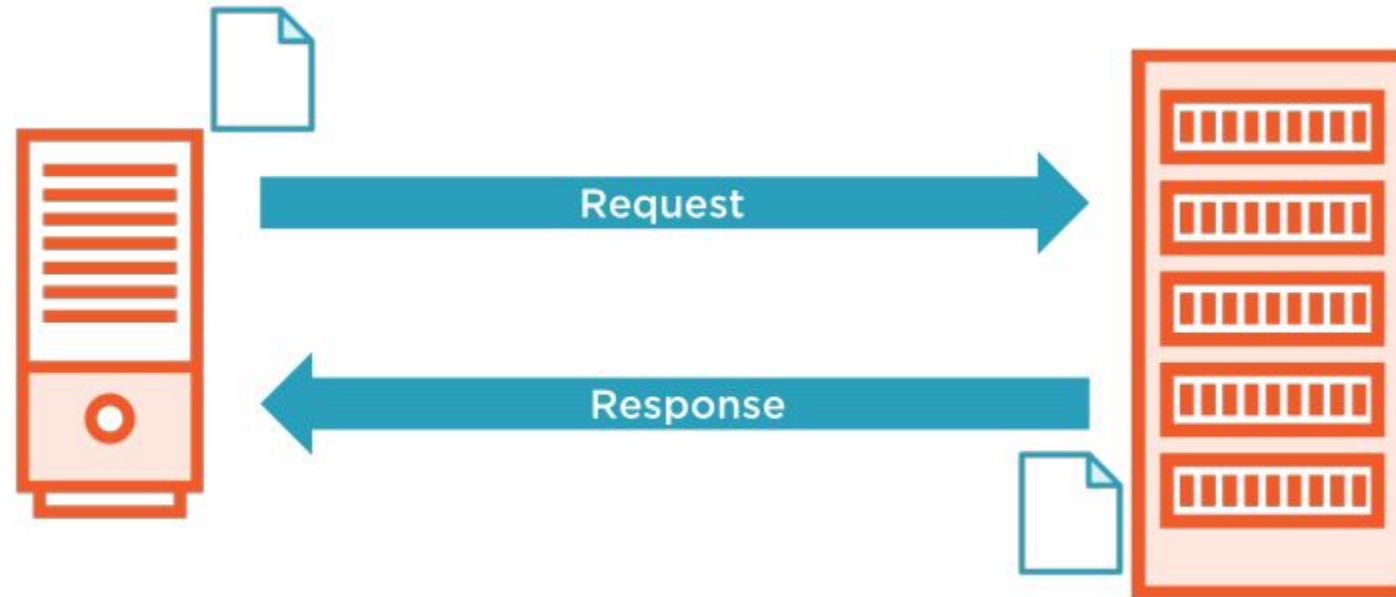
Unary

Server Streaming

Client Streaming

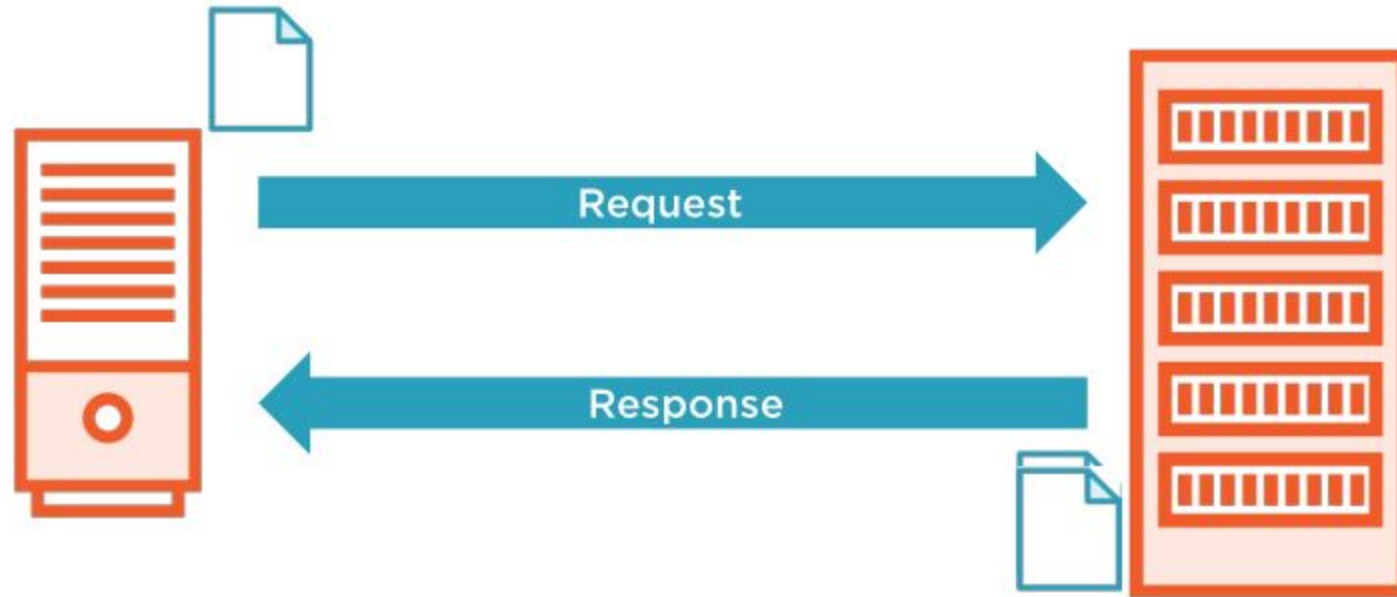
Bidirectional Streaming

Unary RPC



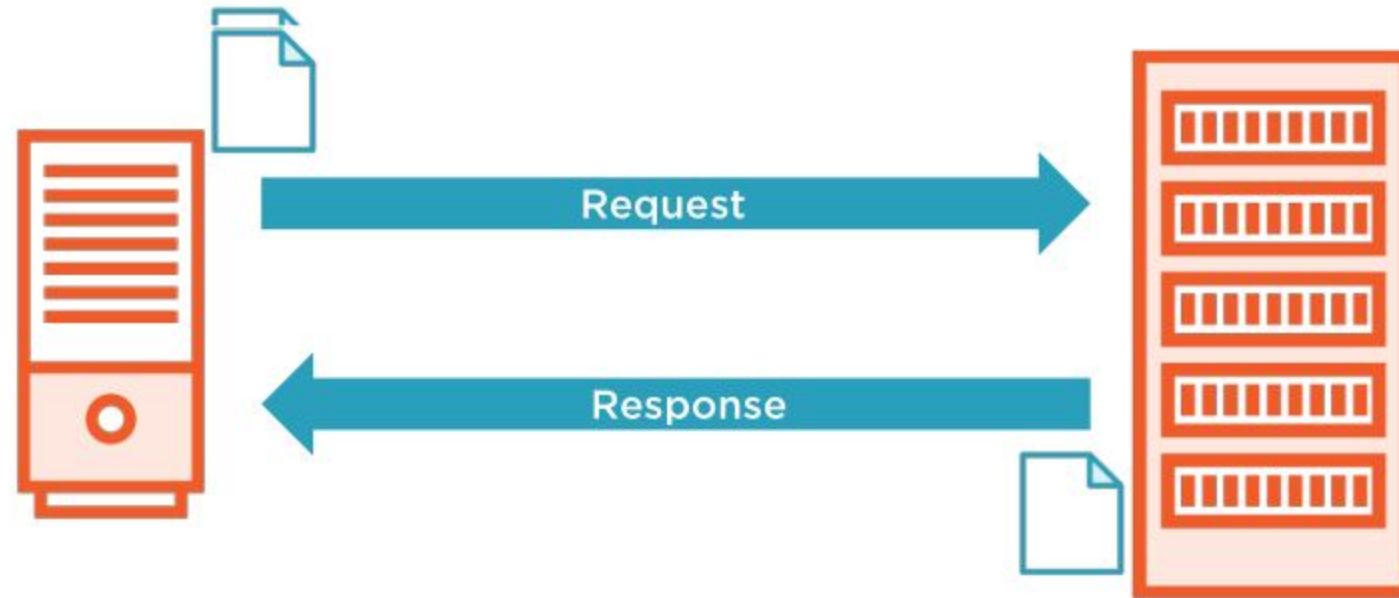
```
rpc Save (EmployeeRequest) returns (EmployeeResponse);
```

Server Streaming RPC



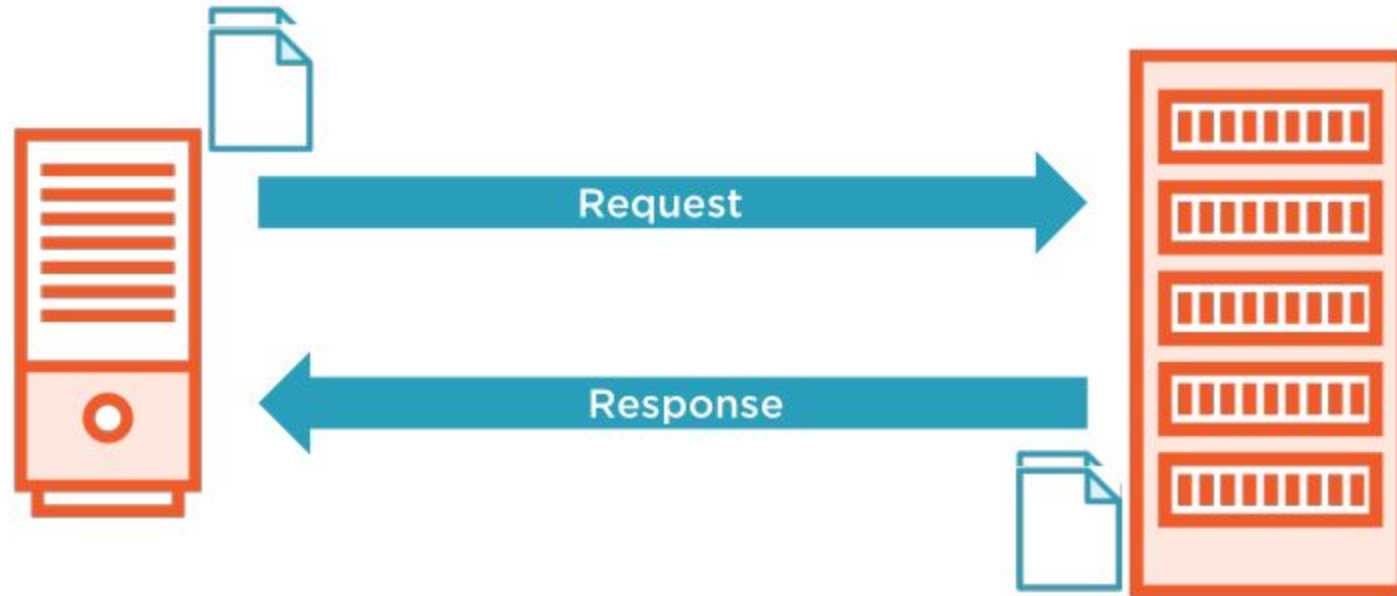
```
rpc GetAll (GetAllRequest) returns (stream EmployeeResponse);
```

Client Streaming RPC



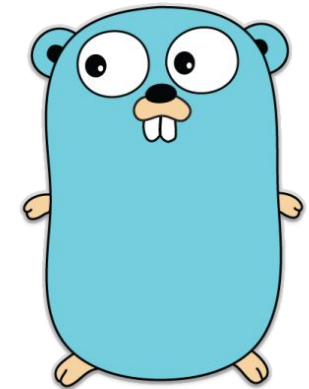
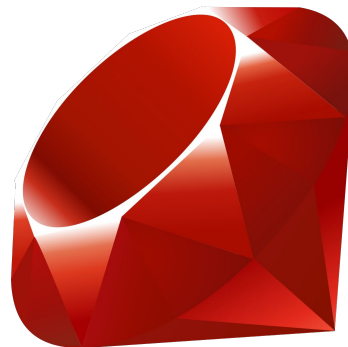
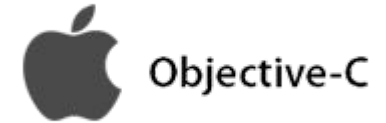
```
rpc AddPhoto (stream AddPhotoRequest) returns (AddPhotoResponse);
```

Bidirectional Streaming RPC

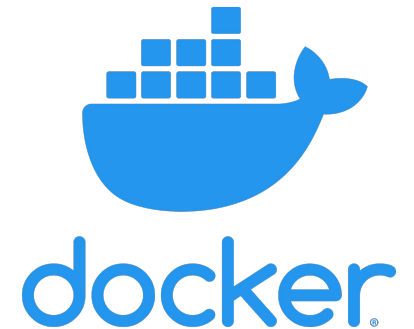


```
rpc SaveAll (stream EmployeeRequest) returns (stream EmployeeResponse);
```

Linguagens de programação suportadas



Principais projetos e companhias que usam gRPC



Awesome gRPC

The screenshot shows the GitHub repository page for `grpc-ecosystem/awesome-grpc`. The repository is described as "A curated list of useful resources for gRPC". It has 237 commits, 23 branches, 0 releases, 48 contributors, and is licensed under CC-BY-4.0. The repository is marked as "awesome" and has 102 watches, 2.2k stars, and 191 forks. The repository is currently on the `master` branch. A pull request #85 by `bojand` is shown, which includes changes to the `LICENSE`, `code-of-conduct.md`, `contributing.md`, and `readme.md` files. The `readme.md` file is displayed, showing the title "Awesome gRPC" with the "awesome" badge, the description "A curated list of useful resources for gRPC", and a "Contents" section with links to "Documentation", "Community", "Official Libraries and Tools", and "Tools" (which includes "CLI" and "GUI").

grpc-ecosystem / awesome-grpc

Watch 102 Unstar 2.2k Fork 191

Code Issues 2 Pull requests 1 Projects 0 Wiki Security Insights

A curated list of useful resources for gRPC

grpc awesome awesome-list protocol-buffers

237 commits 23 branches 0 releases 48 contributors CC-BY-4.0

Branch: master New pull request Create new file Upload files Find file Clone or download

bojand Merge pull request #85 from bufbuild/bufbuild Latest commit 84df3ed 9 days ago

LICENSE	set LICENSE to be CC-BY-4 raw plaintext	2 years ago
code-of-conduct.md	initial commit	2 years ago
contributing.md	fixes	2 years ago
readme.md	Add buf and remove trailing whitespace	9 days ago

readme.md

Awesome gRPC

A curated list of useful resources for gRPC

Contents

- Documentation
- Community
- Official Libraries and Tools
- Tools
 - CLI
 - GUI

<https://github.com/grpc-ecosystem/awesome-grpc>

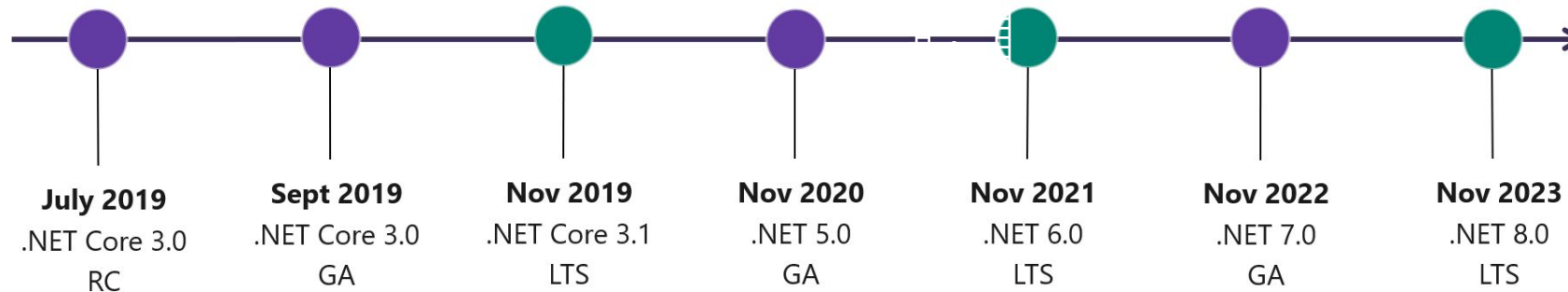
Demo 🍷



Pré-requisitos

Suporte oficial em .NET Core SDK 3.0 lançado em [.NET Conf 2019](#) 23-25 de setembro.
.NET Core 3.1 lançado em 03 de novembro com foco em blazor e desktop.

.NET Schedule



- .NET Core 3.0 release in September
- .NET Core 3.1 = Long Term Support (LTS)
- .NET 5.0 release in November 2020
- Major releases every year, LTS for even numbered releases
- Predictable schedule, minor releases if needed

Pré-requisitos



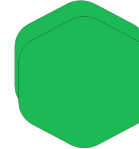
Visual Studio 2019 with the ASP.NET and web development workload



.NET Core SDK 3.0 ou superior



Visual Studio Code



.NET Core SDK 3.0 ou superior



C# for Visual Studio Code (latest version)

Criar um serviço gRPC

```
dotnet new grpc -o GrpcGreeterServer  
code GrpcGreeterServer  
dotnet run  
dotnet dev-certs https --trust
```

Criar o cliente gRPC em um aplicativo de console .NET

```
dotnet new console -o GrpcGreeterClient
code GrpcGreeterClient
dotnet add GrpcGreeterClient.csproj package Grpc.Net.Client
dotnet add GrpcGreeterClient.csproj package Google.Protobuf
dotnet add GrpcGreeterClient.csproj package Grpc.Tools
```

Copiar o arquivo Protos\greet.proto para o GrpcGreeterClient.
Adicionar um grupo de itens no GrpcGreeterClient.csproj.

```
<ItemGroup>
  <Protobuf Include="Protos\greet.proto" GrpcServices="Client" />
</ItemGroup>
```

Criar o cliente Greeter

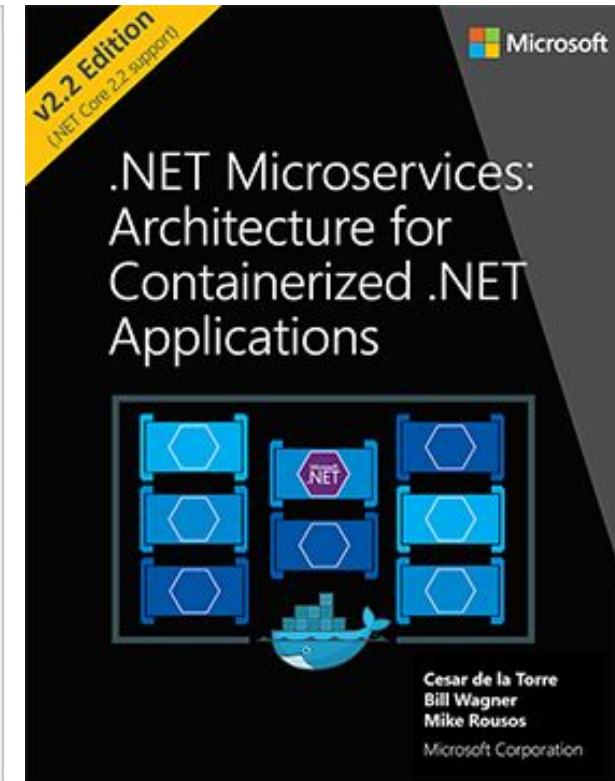
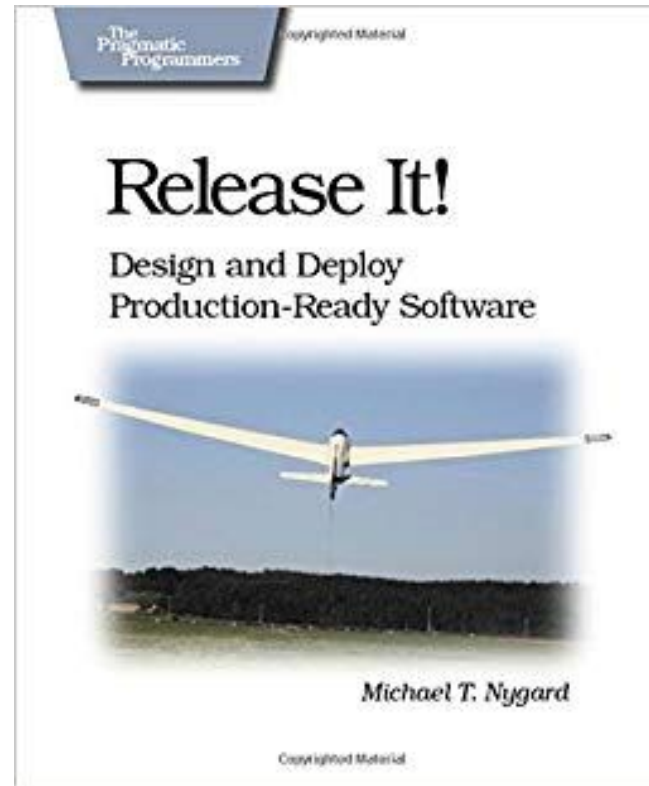
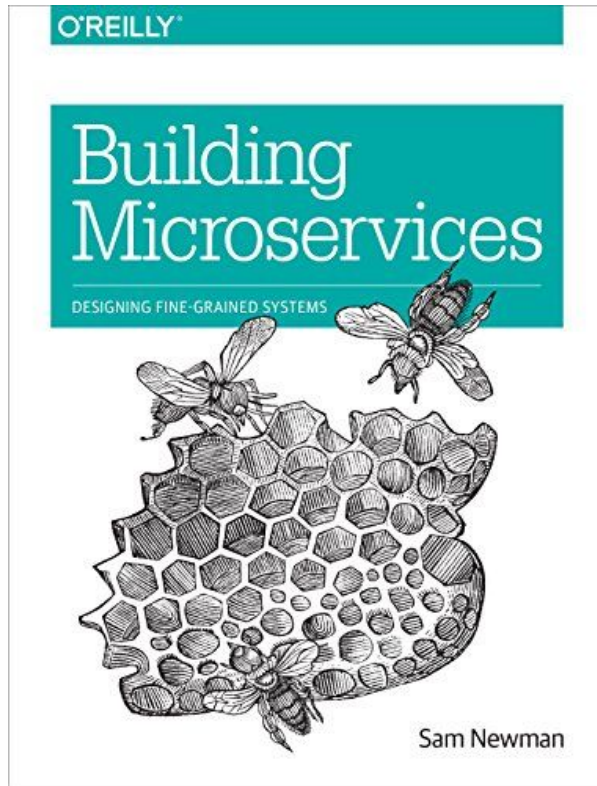
```
using System;
using System.Net.Http;
using System.Threading.Tasks;
using GrpcGreeter;
using Grpc.Net.Client;

namespace GrpcGreeterClient
{
    class Program
    {
        static async Task Main(string[] args)
        {
            // The port number(5001) must match the port of the gRPC server.
            var channel = GrpcChannel.ForAddress("https://localhost:5001");
            var client = new Greeter.GreeterClient(channel);
            var reply = await client.SayHelloAsync(
                new HelloRequest { Name = "GreeterClient" });
            Console.WriteLine("Greeting: " + reply.Message);
            Console.WriteLine("Press any key to exit...");
            Console.ReadKey();
        }
    }
}
```

Referências



Referências para microsserviços



<https://martinfowler.com/articles/microservices.html>

<https://youtu.be/wgdBVIX9ifA> (GOTO 2014 • Microservices • Martin Fowler)

<https://youtube.com/watch?v=CZ3wluvmHeM> (Mastering Chaos - A Netflix Guide to Microservices)

<https://dotnet.microsoft.com/learn/aspnet/microservices-architecture>

<https://www.elemarjr.com/pt/guia-microservicos/>

<https://www.eximia.co/pt/2019/08/21/conways-law/>

<https://www.youtube.com/watch?v=oFpLD3dnSe4> (MongoDB y Microservicios | Power Microservices | Parte I)

<https://www.eximia.co/pt/2019/09/12/relacao-entre-squads-e-microservicos/>

<https://medium.engineering/microservice-architecture-at-medium-9c33805eb74f>

Referências para HTTP/2



Daniel Stenberg

Referências para HTTP/2

<https://www.upwork.com/hiring/development/the-http2-protocol-its-pros-cons-and-how-to-start-using-it/>

<https://stackoverflow.com/questions/28524512/which-web-servers-support-http-2>

<https://developers.google.com/web/fundamentals/performance/http2/?hl=pt-br>

<https://waldyrfelix.com.br/como-funciona-o-http-2-na-pr%C3%A1tica-d71f7ed5ede3>

Referências para Protobuf

<https://developers.google.com/protocol-buffers/>

<https://blog.usejournal.com/what-the-hell-is-protobuf-4aff084c5db4>

Referências para gRPC 🍏

<https://grpc.io/docs/>

<https://docs.microsoft.com/en-us/aspnet/core/grpc/index?view=aspnetcore-3.0>

<https://docs.microsoft.com/en-us/aspnet/core/tutorials/grpc/grpc-start?view=aspnetcore-3.0&tabs=visual-studio-code>

<https://youtu.be/fMq3lpPE3TU> (The Story of Why We Migrate to gRPC and How We Go About It - Matthias Grüter, Spotify)

<https://medium.com/@EdgePress/is-grpc-the-future-of-client-server-communication-b112acf9f365>

<https://www.slideshare.net/mobile/borisoalex/grpc-vs-rest-let-the-battle-begin-81800634>

<https://gago.io/blog/cloud-native-computing-foundation/>

<https://gago.io/blog/grpc-no-asp-net-core-guia-introductorio/>

<https://www.youtube.com/watch?v=rrugn-DDfRI> (gRPC: o novo modelo de comunicação Cross-Platform)

<https://www.bizety.com/2019/04/11/api-battles-grpc-vs-rest/amp/>

<https://app.pluralsight.com/library/courses/grpc-enhancing-application-communication/table-of-contents>

<https://www.bugsnag.com/blog/grpc-and-microservices-architecture>

<https://www.youtube.com/watch?v=Pa6qtu1wls8> (ASP.NET Community Standup - June 4th, 2019 - gRPC!)

<https://www.youtube.com/watch?v=CbNimCiMqe8> (Adopting gRPC at Spotify by Matthias Grüter)



bit.ly/grpc-fb



edinei.dev



github.com/neiesc



linkedin.com/in/neiesc



twitter.com/neiesc



facebook.com/neiesc



t.me/neiesc



OBRIGADO!

